Training and Testing

- Suppose that you have a dataset of 1,000 samples (rows).

- Split it into two groups; one for training and the other for testing.

- The features and labels (inputs and outputs) of the training data are given to the algorithm. (why?)

- For testing, the algorithm is given just the inputs. (why?)

Have a look on the following dataset.

	x1	x2	Y
1	2	4	6
2	7	5	12
3	4	6	11
4	3	9	13
5	11	2	13
6	5	2	9
7	4	3	7
8	8	2	10
9	3	6	10
10	1	2	3

The following rows are randomly selected for training (1,2,4,5,7,9,10).

Training

	x1	x2	Y
1	2	4	6
2	7	5	12
3	4	6	11
4	3	9	13
5	11	2	13
6	5	2	9
7	4	3	7
8	8	2	10
9	3	6	10
10	1	2	3

* Algorithm-A: Y = x1 + x2

* Algorithm-B: Y = x1 + x2 + 1

Testing

	x1	x2	Y
1	2	4	6
2	7	5	12
3	4	6	11
4	3	9	13
5	11	2	13
6	5	2	9
7	4	3	7
8	8	2	10
9	3	6	10
10	1	2	3

Algorithm-A: Y = x1 + x2
row#3: Y = 4+ 6 = 10
row#6: Y = 5 + 2 = 7
row#8: Y = 8 + 2 = 10

Actual value	Algorithm-A value (predicted)
11	10
9	7
10	10

Algorithm-B: Y = x1 + x2 + 1
row#3: Y = 4+ 6 + 1 = 11
row#6: Y = 5 + 2 + 1= 8
row#8: Y = 8 + 2 +1 = 11

Actual value	Algorithm-B value (predicted)
11	11
9	8
10	11

Accuracy of classification

$$Accuracy = \frac{Number\ of\ correct\ predictions}{\text{Total number of predictions}}$$

Example:

Actual value	Algorithm value (predicted)
1	1
1	1
0	1
1	1
0	0
0	1
1	1
0	0
0	0
1	1

Accuracy = 8 / 10 = 80%

Accuracy of regression

$$RMSE = \sqrt{\frac{\sum_{i=1}^{N} (Predicted_i - Actual_i)^2}{N}}$$

Example:

Actual value	Algorithm value (Predicted)
4	6
6	4
8	9
12	12

$\sum = ((6\text{-}4)^2 + (4\text{-}6)^2 + (9\text{-}8)^2 + (12\text{-}12)^2) = (4 + 4 + 1 + 0) = \mathbf{9}$

RMSE $= \sqrt{9/4} = \sqrt{2.25} = \mathbf{1.5}$

Traditional algorithms ~ Machine learning algorithms

Traditional algorithms (e.g. Sorting, search, …)

Inputs are known and the algorithm is specific.

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Supervised M.L algorithms

Inputs and outputs are known (given)

The algorithm will learn (derive the model… find the relation)